

ComicCraft – AI Comic Story Creator

PAGE 1 – PROJECT OVERVIEW & OBJECTIVES

ComicCraft Project

ComicCraft is a web-based AI application designed to generate personalized comic book stories and illustrations from user-provided prompts. The project combines storytelling models and image-generation technology to simplify the comic creation process.

Main Project Objective

The main objective is to allow a user to enter a creative idea and receive a complete comic experience. The system generates a panel-by-panel storyline, narration, character dialogue, comic-style illustrations, and a downloadable PDF.

Key Objectives

- Generate structured comic outlines based on user prompts.
- Generate detailed narration and character dialogues.
- Generate comic-style illustrations for each panel.
- Provide a simple web interface for entering comic details.
- Display the generated comic through a preview page.
- Export the completed comic as a PDF.

Project Scope

The project covers the complete flow from user input to comic export. Users can provide a story prompt, main character name, setting, story tone, and art style. The application processes these inputs through the FastAPI backend and AI services.

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PAGE 2 – USER REQUIREMENTS

User Input Requirements

The ComicCraft application requires several pieces of information from the user. These inputs help the system create a personalized comic that matches the user's creative idea.

1. Story Prompt

The story prompt represents the main idea of the comic. For example, a user may describe a story about a brave fox exploring an enchanted forest.

2. Main Character Name

The user can provide the name of the main character. This information is used to personalize the generated story and dialogue.

3. Setting

The setting specifies where the story takes place. Examples described in the project include a forest, school, city, or other locations.

4. Story Tone

The user can select a tone such as dramatic, funny, light-hearted, or poetic. The selected tone influences the generated narrative.

5. Art Style

The user can select a preferred visual style, such as anime, comic book, pixel art, or realistic.

Expected User Experience

The interface is intended to be simple and user-friendly. The user enters the required information, submits the form, views the generated comic panel-by-panel, and can then download the final comic as a PDF.

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PAGE 3 – FUNCTIONAL REQUIREMENTS

Core Functional Requirements

The project requirements include several connected functions that together form the comic-generation pipeline.

Story Outline Generation

The system must generate a structured 5-panel comic outline. Each panel contains a panel number, title, scene description, and image-generation prompt.

Story & Dialogue Generation

The system must expand the panel outline into a complete comic-style story containing narration and character dialogues for each panel.

Image Generation

The system must create a comic-style illustration for each panel using the image-generation prompt produced during the story workflow.

Layout Building

The generated images and story content must be organized into a structured comic layout. Each image is matched with its corresponding panel story.

PDF Export

The application must compile the comic panels, images, and narration into a multi-page PDF. The project uses FPDF for PDF generation.

Preview & Download

Users must be able to preview the generated comic through the web interface and download the completed comic as a PDF.

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PAGE 4 – TECHNICAL & SYSTEM REQUIREMENTS

Technology Requirements

The project requires Python-based development and several supporting frameworks and AI libraries.

Backend Requirements

FastAPI is used to manage server-side operations, routing, form data, JSON requests, AI workflow coordination, and template rendering.

AI Requirements

Gemini Flash is selected for fast, structured comic panel outlines. Gemini Pro is selected for detailed narration and character dialogue. Stable Diffusion is selected for comic-style illustrations.

Frontend Requirements

The frontend uses HTML, CSS, and Jinja2 templates. The main templates include index.html for input, comic_preview.html for the generated comic, and export_success.html for export confirmation.

Development Environment

The project requires Python and pip. A virtual environment is recommended to isolate project dependencies. Required libraries include FastAPI, Uvicorn, Jinja2, Google Generative AI, Diffusers, Transformers, FPDF, Pillow, and related packages.

API & Security Requirements

Sensitive information such as the Gemini API key and Hugging Face API token should be stored using environment variables rather than directly exposing them in application code.

Version Control

Git is included in the project's prerequisites for version control and project management.

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PAGE 5 – REQUIREMENTS SUMMARY & PLANNING CHECKLIST

Complete Requirement Summary

ComicCraft is planned as an end-to-end AI comic creation system. The planning stage identifies what the application should accept, generate, display, and export.

Input Requirements

- Story prompt
- Main character name
- Setting
- Story tone
- Art style

Processing Requirements

- Generate a structured 5-panel outline.
- Generate narration and character dialogue.
- Generate comic-style images.
- Organize images and story content into panels.
- Create a downloadable PDF.

Output Requirements

The expected output is a complete personalized comic containing panel titles, illustrations, scene descriptions, captions or narration, and a downloadable PDF version.

Planning Checklist

- Understand project objectives.
- Identify user inputs and expected outputs.
- Select suitable AI models.
- Define frontend, backend, and AI integration components.
- Identify required Python libraries and development tools.
- Plan the comic-generation workflow.
- Plan preview and PDF export functionality.

- Prepare the application for local testing and deployment.

Conclusion

The Project Planning & Requirements phase establishes the foundation for ComicCraft. By clearly defining the user inputs, system functions, technologies, and expected outputs, the project can proceed systematically toward AI comic generation, preview, and PDF export.